

Mini–Capstone Project: TDOA Localization of Fast Radio Bursts

Author: Prof. Mohit Bhardwaj

Project Deliverable: A single, clean Jupyter Notebook containing well–documented Python code (using `numpy`, `matplotlib`, and optionally `astropy`), all required plots, brief explanations in Markdown, and answers to comprehension questions.

1 Background and Physical Motivation

Fast Radio Bursts (FRBs) are bright, millisecond–duration radio transients originating at extragalactic distances. Because they are so short–lived, precise time–stamping of the burst arrival is essential. At frequencies around 1–2 GHz, individual radio dishes typically have very large beams (hundreds to thousands of deg^2), and thus a single telescope alone cannot localize an FRB with useful precision.

1.1 Geometric Time Delays and TDOA

Consider a plane electromagnetic wave arriving from direction $\hat{\mathbf{s}}$, a unit vector in a local coordinate system. Let $\mathbf{r}_i$ denote the position of telescope i in meters. The arrival time at that telescope is

$$t_i = \frac{\mathbf{r}_i \cdot \hat{\mathbf{s}}}{c}, \quad (1)$$

where c is the speed of light. Absolute arrival times are not needed: only the *time differences of arrival* (TDOA)

$$\Delta t_{i0} = t_i - t_0 = \frac{(\mathbf{r}_i - \mathbf{r}_0) \cdot \hat{\mathbf{s}}}{c}. \quad (2)$$

TDOA encodes information about the source direction. If timing precision is σ_t , then the corresponding uncertainty in geometric path length is $c\sigma_t$. For example, a $10\,\mu\text{s}$ timing uncertainty corresponds to $\sim 3\,\text{km}$ uncertainty in projected path difference.

1.2 The Local ENU Coordinate System

We adopt the standard East–North–Up (ENU) coordinate system. Each telescope position is

$$\mathbf{r}_i = (x_i, y_i, z_i), \quad z_i \approx 0 \text{ (flat ground assumption)}.$$

A source with altitude (elevation) alt and azimuth az corresponds to a unit vector

$$\hat{\mathbf{s}} = \begin{pmatrix} \cos(\text{alt}) \sin(\text{az}) \\ \cos(\text{alt}) \cos(\text{az}) \\ \sin(\text{alt}) \end{pmatrix}. \quad (3)$$

This relation allows you to map any point on the visible hemisphere into a direction vector for computing predicted time delays.

1.3 Why Two Telescopes Yield a “Ring”

With two telescopes (one baseline), the single TDOA equation

$$\Delta t_{10} = \frac{\mathbf{b}_{10} \cdot \hat{\mathbf{s}}}{c}$$

defines all directions whose dot–product with the baseline $\mathbf{b}_{10} = \mathbf{r}_1 - \mathbf{r}_0$ produces the same geometric delay. This locus is a “ring” or annulus on the sky. Therefore,

Two telescopes cannot uniquely localize an FRB. They restrict the source to a ring.

1.4 Why Three Telescopes Localize a Source

With three non-collinear telescopes, two independent baselines provide two independent constraints:

$$\Delta t_{10}, \Delta t_{20}.$$

The intersection of their corresponding rings reduces to a small two-dimensional region in (alt, az). This region—when computed statistically—defines your localization uncertainty.

1.5 Using χ^2 to Estimate Localization Regions

Your notebook will estimate the most likely sky position using a grid-based χ^2 statistic:

$$\chi^2(\text{alt}, \text{az}) = \sum_i \frac{(\Delta t_{i0}^{\text{pred}} - \Delta t_{i0}^{\text{obs}})^2}{\sigma_t^2}. \quad (4)$$

Here $\Delta t_{i0}^{\text{obs}}$ are the “measured” TDOAs (you will simulate these), and $\Delta t_{i0}^{\text{pred}}$ are predicted TDOAs for each grid point. The best-fit direction minimizes χ^2 . The 68% (1σ) confidence region for two fitted parameters (alt and az) is given by

$$\Delta\chi^2 = \chi^2 - \chi_{\min}^2 \leq 2.3.$$

This “ $\Delta\chi^2$ contour” defines your localization patch.

2 Project Tasks and Requirements

Your task is to implement an end-to-end localization simulation in a single Jupyter Notebook. Write all explanatory text in Markdown cells and all numerical work in Python.

2.1 Section 2: Direction Vectors

- Implement a function `altaz_to_unit_vector(alt, az)`.
- Verify correctness by plotting unit vectors for a set of altitudes and azimuths.

2.2 Section 3: Two-Antenna TDOA Localization

Assume a baseline length of approximately 15 km between the two stations.

- Choose a true FRB direction (alt, az).
- Compute true geometric delays and add Gaussian timing noise of $10\mu\text{s}$.
- Compute a 2D χ^2 map (alt-az grid) based on a single baseline.
- Plot the resulting **ring-shaped** localization band (the 1σ region).

Comprehension Questions:

1. Why does a single baseline restrict the FRB only to a ring?
2. How would the ring *width and curvature* change if the baseline length were decreased from 15 km to 5 km? Explain using the delay equation.
3. How would the ring change if the baseline were increased beyond 15 km?

2.3 Section 4: Three-Antenna Localization

Add a third station such that the three antennas are not colinear.

- Compute TDOAs for all baselines.
- Produce a 2D χ^2 heatmap over (alt, az).
- Identify the minimum- χ^2 location.
- Plot the 1σ ($\Delta\chi^2 = 2.3$) localization contour.
- Compute the localization area in deg^2 (using $\cos(\text{alt}) d\text{alt} d\text{az}$).

Comprehension Questions:

1. Why do three stations collapse the ring into a compact patch?
2. How does the shape of this patch depend on the geometry of the triangle formed by the stations?
3. Briefly explain why intensity-only data (not voltages) is sufficient for TDOA localization.

2.4 Section 5: Localization as a Function of Sky Position

- Compute the 1σ localization area for a grid of FRB positions over altitudes 20° – 85° and azimuths 0° – 360° .
- Produce a 2D heatmap of $\log_{10}(\text{localization area})$.

Comprehension Questions:

1. Why does localization degrade at low altitudes?
2. Do certain azimuths yield worse localization for your chosen array? Why?

2.5 Optional (Star Category): Four or More Stations

For students who wish to explore further:

- Add a fourth (or fifth) station within a ≤ 15 km radius of the center.
- Investigate how the localization area scales with the number of stations.

3 Final Deliverables

Your submitted Jupyter Notebook must contain:

- All Python code with informative comments.
- All required plots (saved as PNGs in a **figures/** folder).
- Markdown explanations of each step.
- Answers to all comprehension questions.
- A brief concluding paragraph summarizing what you learned.

End of Assignment